Inference on the Effects Size of Violent Crame Rates on PTSD Prevalence in Chicago

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Post-Traumatic Stress Disorder (PTSD) is a complex psychiatric disorder which occurs in those who have experienced traumatic events and is characterized by a variety of debilitating symptoms including intrusive memories or nightmares, avoidance behaviors, and negative changes in mood and cognition. Prior research has concluded that the prevalence of PTSD on the south side of Chicago is significantly higher than in the general population and suggests that there are conditions which may work to produce this heightened prevalence.

While the south side of Chicago is known for its violence and systematic socioeconomic deprivation, the significance to which violence may contribute to these phenomena is unclear. This analysis aims to make inference of the relative significance and effects size of violence exposure on the prevalence of PTSD controlling for relevant factors.

The Mental Health America's 2020-2023 zip code level statistics for positive survey responses indicating PTSD were collected for every zip code in Chicago as they were not available on a neighborhood level. From the Chicago Health Atlas, the 2020 statistics for violent crime incidence, poverty level, demographic, healthcare, and economic variables were collected for each zip code. While health outcomes in Chicago have very strong geographic correlations, it is significant to note that zip codes do not correspond as cleanly to neighborhood and community boundaries which may obscure relevant systematic socioeconomic patterns because they are drawn for logistic purposes and postal address density. Naturally, as the south and west sides of Chicago have lower population densities, zip codes in these portions of the city are larger as a result and require caution when interpreting geographic patterns in these areas. There were a resulting ($n = 58$) zip codes included which unfortunately does not result in high statistical power.

For each zip code, PTSD prevalence, violent crime, and drug crime rate in proportion to the zip code level population were calculated. The summary statistics for each of these variations were calculated and the distributions were analyzed. Most of the variates' distributions were not approximately normal except for house cost burden, population, and positive PTSD cases. Variates associated with hardship including poverty, unemployment rate, food insecurity rate, violent crime rate, drug crime rate, and PTSD prevalence had multi-modal and strongly positive skewed distributions where zip codes on the south and west sides skewed higher. These distributions correspond to repeatedly observed patterns in urban settings where communities can be divided by a two-tier system often corresponding to racial division.

To assess the relationships between these socioeconomic markers, Pearson, Kendall, and Spearman correlations were computed between these variates to assess the nature and strength of these correlations. The strongest positive correlations with PTSD prevalence across correlation methods were location affordability index, percent white, high school graduation rate, and prevalence of active transport to work. The strongest negative correlations were Medicaid coverage, unemployment rate, poverty rate, food insecurity rate, and violent crime rate. Of these, High school graduation rate and percent white had high positive correlations along with drug crime and violent crime. Population had a negative correlation with high school graduation rate while high school graduation rate had a strong negative correlation with percent white. These relationships were taken into account during modeling in consideration of collinearity.

Considering the extreme levels of collinearity in the data, the relationships were initially assessed

independently with Poisson generalized linear models fit for each variable with high correlations with PTSD prevalence. The count of positive survey responses for PTSD was used as the outcome variable with an offset for population. After model diagnostics it was assessed that there was over dispersion which made Poisson an inappropriate distribution to model PTSD counts so the models were re-specified with negative binomial. The models which had resulting significant were violent crime rate, housing cost burden, location affordability index, percent who take active transport to work, foreign born, and percent white. Surprisingly, violent crime and housing cost burden had negative parameter estimates while location affordability index, percent who take active transport to work, foreign born, and percent white had positive parameter estimates.

This is not expected as according to prior research; immigrants do not commit crimes as frequently as natural born American citizens and there is no reason to believe that white Americans are more likely to have PTSD than any other group (Roberts et al.). This raises suspicions as to the quality of the data like the representatives of the sampling or response bias from minority groups that this might not be answering honestly due to suspicions of the healthcare system or medical research due to past racial injustices (Richardson et al.).

To control for socioeconomic factors and sampling bias, a new model was specified using violent crime rate, poverty rate, percent white, and percent of population using active transportation as predictors for positive PTSD cases, which was modeled using the negative binomial distribution to account for dispersion. The resulting model showed that violent crime rate (IRR $= 1.20$, $p = .003$) and active transportation rate (IRR $= 1.14$, $p = .005$) were associated with elevated rates of positive cases, while poverty rate was associated with lower rates (IRR $= 0.77$, $p < .001$), where in contrast, percent White was no longer significant ($p = .388$). The residual deviance (70.32) was higher than residual degrees of freedom (53), which suggests that the negative binomial model is still not accounting for some component of the data, which could likely be spatial correlations considering that these socioeconomic variables often correlate strongly on a spatial basis. The dispersion parameter was ($\theta = 19.72$)

A negative binomial random effects model was fit to account for this assuming that the variation was random and not spatially correlated. This method was used as a tool to regularize extreme observations rather than to model cluster specific effects as it is usually used for. The resulting model shows that controlling for included covariates, violent crime rate (IRR $= 1.35$, $p < .001$), percent foreign born (IRR $= 1.11$, $p = .009$), and percent white (IRR $= 1.21$, $p = .002$) were associated with elevated rates of PTSD prevalence, while poverty rate was associated with lower rates (IRR $= 0.81$, $p = .002$). The random intercept for zip code ($\sigma = 0.23$) suggests there is in fact variability between zip codes, indicating that there are area varying factors that contribute to differences in PTSD prevalence. The dispersion parameter was ($\theta = 2.44e + 07$) which shows that these unobserved differences may in fact be spatially autocorrelated, as previously suspected as the random intercept accounted for most of the dispersion in the first model. However, as a random intercept was fit for every zip code, which extremely limits the inferential use of this model's random intercepts. It is noteworthy that the AIC of the model without the random intercept had an AIC 525.36 of while the random intercept model's AIC was 526.2 which shows their performance is nearly identical which was expected due to the nature of there being no repeated measures within clusters.

To further assess the evidence of their being spatial autocorrelation, a hierarchical cluster analysis was performed on housing cost burden, location affordability index, active transport, percent foreign born, poverty rate, unemployment rate, high school graduation rate, food insecurity rate, Medicaid coverage, violent crime rate, and drug crime rate. Count variables including population were not included due to the magnitude of variance characteristic of the between zip code population counts and crime incidence. Within cluster sum of squares method was used to decide the optimal number of clusters where (k = 4) was chosen. The spatial cluster membership of the zip codes was visually assessed and there was a very clear spatial pattern as to the distribution of the clusters where downtown all belonged to the same cluster and one cluster dominating the west and south sides, which corresponded to areas with high socioeconomic distress (Figure 1). This provided further evidence of spatial autocorrelation, which may have been the source of the high dispersion in the initial models.

The low sample size (n = 58) severely limits the power and compatibility of frequentist methods, so a

Bayesian CAR model model was specified with predictors as violent crime rate, poverty rate, high school graduation rate, and percent white with an offset for population. The spatial structure of Chicago's zip codes was implemented with a Besag–York–Mollie conditional autoregressive prior. Weakly informative priors were set in accordance to Wang et al. and were assigned to all parameters with normally distributed priors for regression coefficients, and exponential priors for the PTSD positive cases and the BYM2 component. The model was implemented in STAN using HMC with four chains and 1000 iterations per chain, with parameters adjusted for stability in its convergence. The resulting model found that violent crime rate (IRR = 1.32, 95% CrI [1.13, 1.57]) were associated with elevated PTSD prevalence in contrast to poverty rate being associated with lower PTSD prevalence (IRR = 0.78, 95% CrI [0.64, 0.94]). High school graduation rate (95% CrI [0.86, 1.25]) and percent white (95% CrI [0.91, 1.48]) were not greatly associated with PTSD prevalence. The model's spatial correlation parameter ($\rho = 0.55$, 95% CrI [0.03, 0.98]) suggested that there was moderate spatial dependence between neighboring zip codes although there was considerable uncertainty with the spatial variance component ($\sigma = 0.10$, 95% CrI [0.00, 0.27]) suggesting that there is some spatial heterogeneity. The estimated dispersion parameter (shape = 8.77) suggests that overdispersion was less prevalent than in the prior models while still being present, so this model was a significant improvement in comparison.

Throughout the specified regression models, PTSD prevalence has had a strange suggested relationship with racial demographics and poverty rate as which earlier discussed, points to either some systematic sampling bias, or model misspecification. To further investigate this, a structural equation model was specified for the proposed mediation pathway where violent crime mediates the relationship between poverty levels and PTSD prevalence. A model was specified for violent crime rate which included poverty rate, high school graduation rate, and percent white as predictors approximated with the Gamma distribution. The model for positive PTSD cases was specified with violent crime rate, poverty rate, high school graduation rate, and percent white as predictors with PTSD cases approximated with the negative binomial distribution. Both included BYM2 components to model the spatial dependency. Like the previous model, weakly informative priors were used. The model was likewise implemented in STAN using HMC with four chains and 1000 iterations per chain, with parameters adjusted for stability in its convergence.

The resulting model found that violent crime rate (IRR = 1.32, 95% CrI [1.13, 1.57]) was associated with elevated PTSD prevalence while poverty rate was associated with lower PTSD prevalence directly ($\text{IRR}_{\text{direct}} = 0.77$, 95% CrI [0.63, 0.94]). Considering the mediation structure, elevated poverty was associated with elevated violent crime (IRR = 2.08, 95% CrI [1.14, 3.66]), which was associated with elevated PTSD counts, indicating a positive indirect effect by the way of violent crime ($\text{IRR}_{\text{indirect}} = 1.24$). The resulting total effect of poverty on PTSD was near zero ($\text{IRR}_{\text{total}} = 0.95$). High school graduation rate (95% CrI [0.86, 1.25]) and percent white (95% CrI [0.91, 1.48]) showed no clear association with PTSD prevalence neither as a direct or indirect effect. The model's spatial correlation parameter ($\rho = 0.55$, 95% CrI [0.03, 0.98]) suggested moderate spatial dependence between neighboring ZIP codes but the spatial variance component ($\sigma = 0.10$, 95% CrI [0.00, 0.27]) suggested uncertainty with regards to residual spatial heterogeneity. The estimated dispersion parameter ($\theta = 8.77$) suggested that overdispersion was present but even less pronounced than in the previously specified models. Residuals were analyzed with Moran's I across all ZIP codes and high residuals were primarily associated with the Chicago loop area (Figure 3).

Results

At a 95% confidence level, violent crime rate acted as a primary mediator for the relationship between poverty and PTSD prevalence. Results from the Bayesian Structural Equation Model indicate that for every one standard deviation increase in the normalized, cube-root-transformed violent crime rate, PTSD prevalence is predicted to increase by 34% (IRR = 1.34, 95% CI [1.14, 1.57]).

Poverty exerted a strong indirect effect on PTSD through its association with violent crime (Path A IRR = 2.08; Indirect IRR = 1.24). However, the direct effect of poverty on PTSD was negative (IRR = 0.77), resulting in a non-significant total effect ($\text{IRR}_{\text{total}} = 0.95$). This suggests that in Chicago, the impact of poverty on prevalence of PTSD is primarily through the exposure to violent crime rather than poverty in isolation.

Discussion

While the results are considerably robust, the unreliability of the survey data and the inconsistency in the coverage of research on this topic makes interpreting the results difficult. There were many instances where it was clear that the nature of the data made it very difficult to use traditional statistical methods due to the extreme variation in the data. The zip codes 60604, 60606, 60629, 60654 which roughly correspond to the Loop, Marquette Park, Chicago Lawn, the Near North Side, and River North continued to be troublesome throughout the analysis. Each of these communities has unique socioeconomic conditions which make them stand out in the context of the city including extremely high prosperity in the case of the Loop and River North, while Chicago Lawn and Marquette Park had outstandingly bad socioeconomic makers.

Despite how Mental Health America frames its methods, there are clear and systematic sampling and representation issues evident throughout the course of the analysis including areas with higher white populations having higher reported PTSD prevalence in comparison to predominantly black and Latino neighborhoods which have previously been noted to have remarkably high rates of PTSD having lower prevalence of PTSD in comparison. Regardless of there certainly being no ill intention, these survey results paint harmful and dismissive picture of the impacts of redlining, systematic injustice, and community violence in these communities.

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Appendix

Summary Statistics and Models

Variable	**Summary Statistics** M ± SD[1]
Housing Cost Burden	38.14 ± 7.70
Location Affordability Index	48.07 ± 3.38
Percent of Workers Commuting by Active Transportation	25.96 ± 13.07
Percent Foreign Born	20.02 ± 10.97
Percent White	35.85 ± 25.47
Population	47,592.48 ± 24,630.38
Poverty Rate	16.24 ± 8.75
Unemployment Rate	8.49 ± 4.98
High School Graduation Rate	89.07 ± 8.23
Food Insecurity Rate	14.87 ± 4.62
Violent Crime Incidences	457.66 ± 408.19
Drug Crime Incidences	101.78 ± 187.58
medicaid_coverage	22.75 ± 14.86
Violent Crime Rate	0.01 ± 0.01
Drug Crime Rate	0.00 ± 0.01
Positive Responses Indicating PTSD	83.10 ± 44.76
PTSD Rate	0.00 ± 0.00

[1] Mean ± SD

Correlations with PTSD Prevalence

Variable	Pearson (r)	Spearman (ρ)	Kendall (τ)
Location Affordability Index	0.765	0.509	0.371
Percent of Workers Commuting by Active Transportation	0.566	0.390	0.239
Population	-0.546	-0.312	-0.220
High School Graduation Rate	0.406	0.603	0.425
Violent Crime Incidences	-0.381	-0.563	-0.401
Drug Crime Rate	0.362	0.016	0.002
Positive Responses Indicating PTSD	-0.360	0.040	0.040
Percent White	0.353	0.681	0.488
Food Insecurity Rate	-0.318	-0.580	-0.407
Poverty Rate	-0.309	-0.578	-0.405
Violent Crime Rate	0.298	-0.189	-0.117
Unemployment Rate	-0.275	-0.604	-0.440
Percent Foreign Born	0.213	0.213	0.136
Drug Crime Incidences	-0.178	-0.326	-0.222
Housing Cost Burden	0.033	-0.447	-0.321

Negative Binomial Model With Controls

Predictor	IRR	Lower 95% CI	Upper 95% CI	p
Violent Crime Rate	1.205	1.072	1.351	0.002583571
Poverty Rate	0.772	0.678	0.879	1.77×10^{-4}
Percent White	1.061	0.928	1.212	0.387528985
Prop. Commuting by Active Transportation	1.142	1.041	1.253	0.004514252

Bayesian Negative Binomial CAR Model

Predictor	IRR	Lower 95% CrI	Upper 95% CrI
Violent Crime Rate	1.324	1.125	1.564
Poverty Rate	0.777	0.639	0.941
High School Graduation Rate	1.033	0.861	1.240
Percent White	1.161	0.917	1.478

Bayesian CAR Structural Equation Model

Mediation Path	IRR	Lower 95% CrI	Upper 95% CrI
Poverty Rate →Violent Crime	2.065	1.138	3.661
High School Graduation Rate →Violent Crime	1.581	0.890	2.763
Percent White →Violent Crime	0.533	0.258	1.135
Violent Crime Rate →Violent Crime	1.330	1.133	1.569
Poverty Rate →Violent Crime	0.772	0.634	0.938
High School Graduation Rate →Violent Crime	1.024	0.842	1.241
Percent White →Violent Crime	1.165	0.901	1.514

Figures

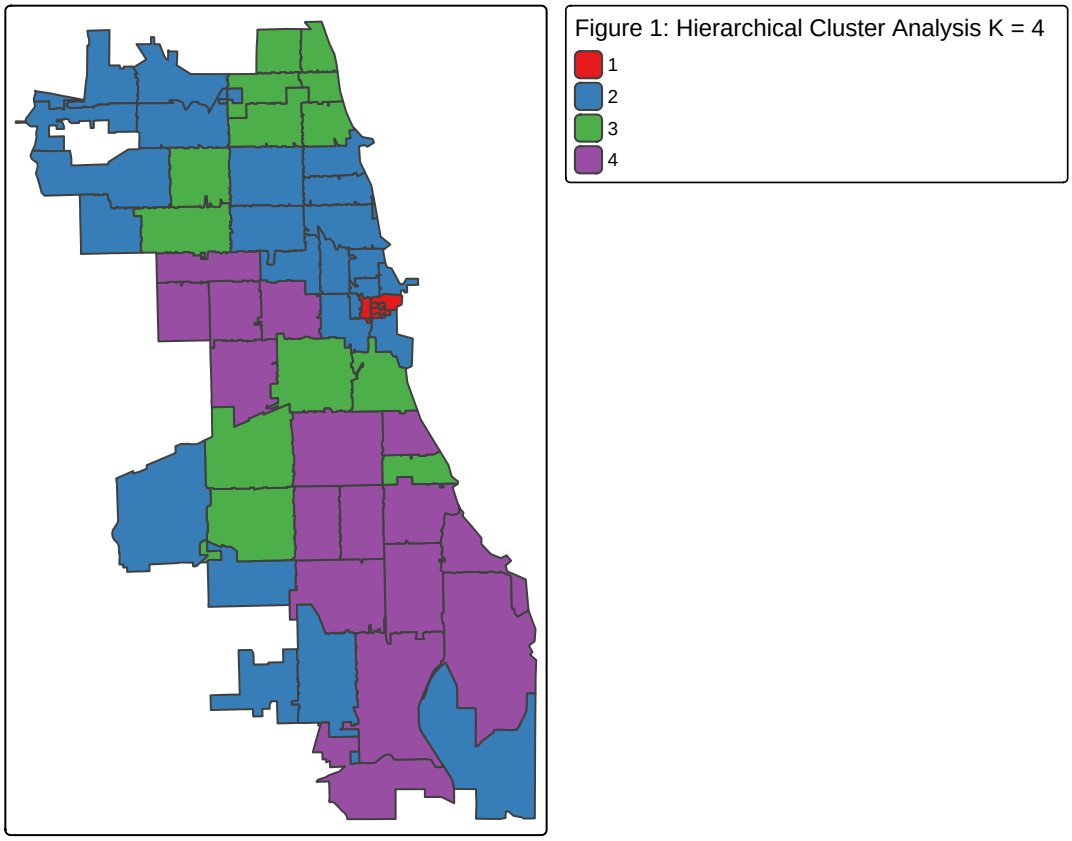
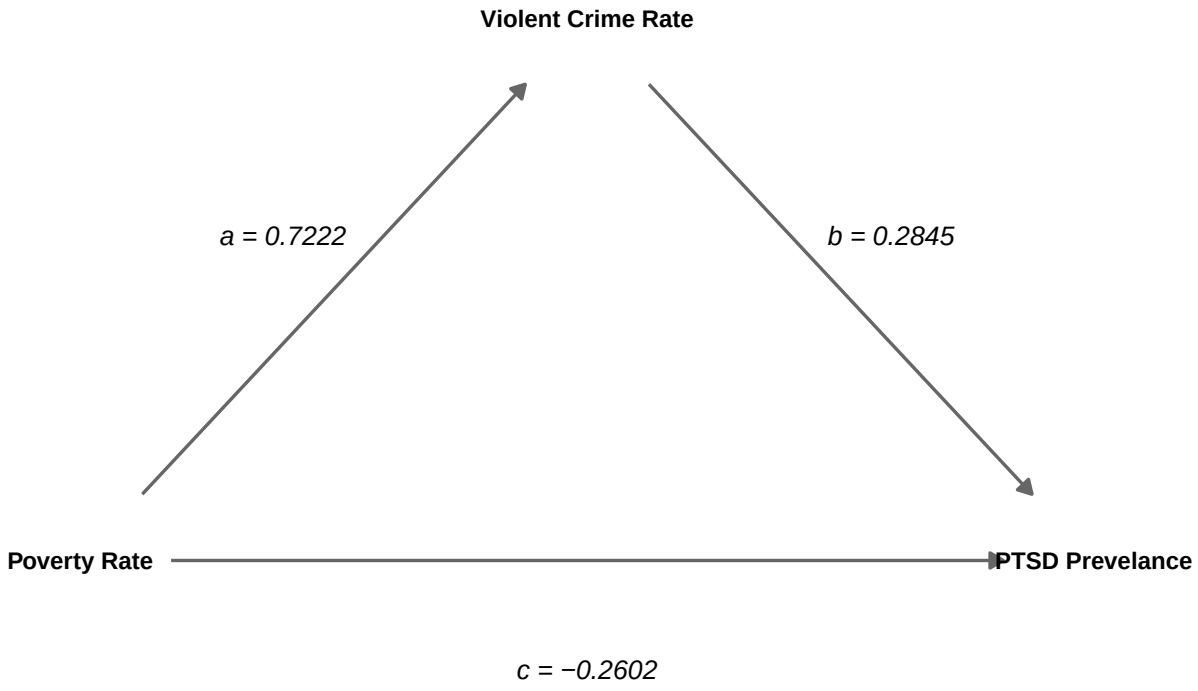


Figure 2: Bayesian SEM CAR Model



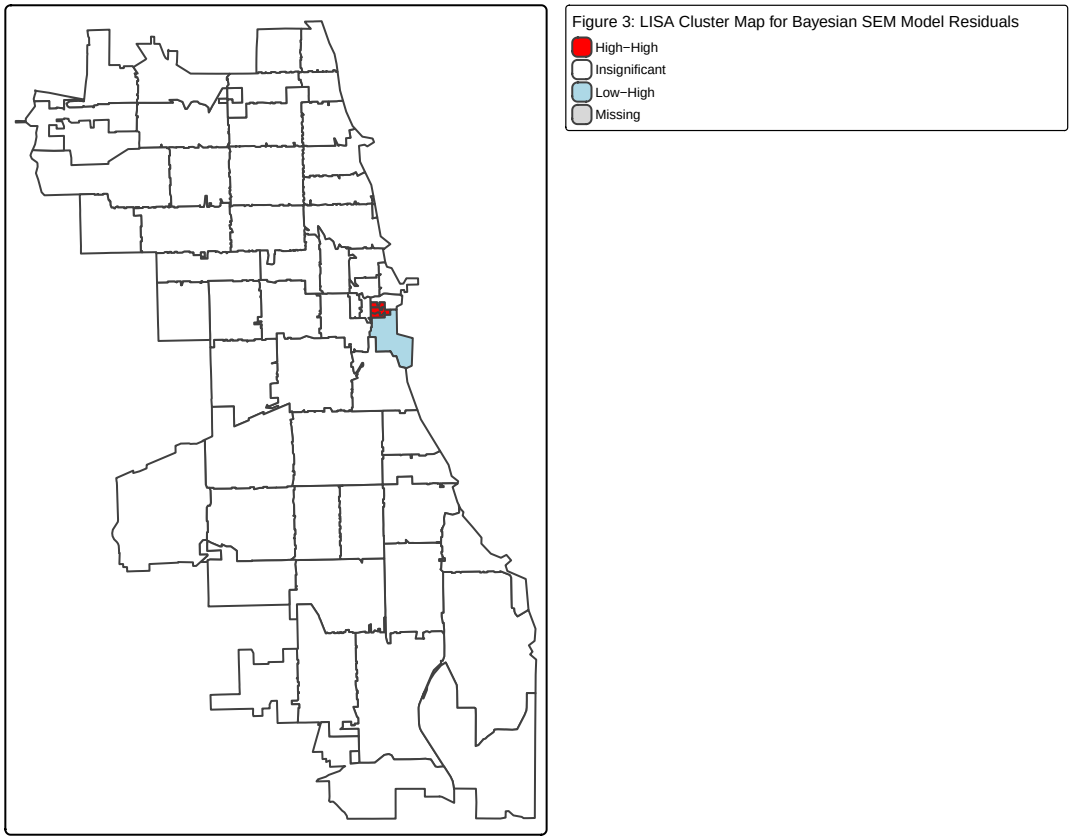


Figure 3: LISA Cluster Map for Bayesian SEM Model Residuals
High−High
Insignificant
Low−High
Missing